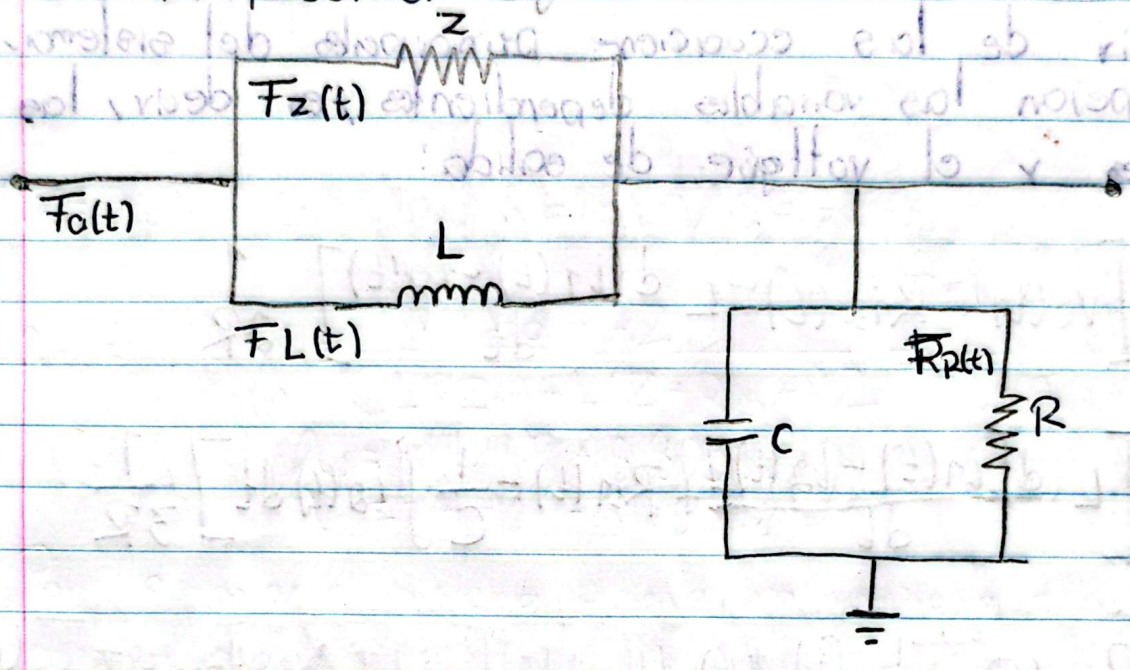
Modelado de sistema Fisiológico

D 07 M 10 A 26

Tercera practica



$$F_a(t) = F_z(t) + F_L(t) = F_c(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z} \qquad F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_c(t) = C\frac{dP_p(t)}{dt} \qquad F_r(t) = \frac{P_p(t)}{R}$$

Transformada de Laplac

$$F_o(s) = F_z(s) + F_L(s) = F_c(s) + F_R(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{Z} \qquad F_L(t) = \frac{1}{L}\int [P_o(s) - P_p(s)]\,dt$$

$$F_c(s) = C\frac{dP_p(s)}{d} \qquad F_c = C_s P_p(s)$$

$$F_R(t) = \frac{P_p(s)}{R}$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{L_s} = C_s P_p(s) + \frac{P_p(s)}{R}$$

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Procedimiento algebraico

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{L} - \frac{P_p(s)}{L} = C_s P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{L_s}\right) P_a(s) = \left(C_s + \frac{1}{R} + \frac{1}{Z} + \frac{1}{L_s}\right) P_p(s)$$

$$\frac{L_s + Z}{LZ_s} P_a(s) = \frac{CLRZs^2 + LZ_s + LR_s}{LRZ_s} P_p(s)$$

$$\frac{L_s + Z}{LZ_s} P_a(s) = \frac{CLRZ_s^2 + L(Z+R)s + RZ}{LRZ_s} P_p(s)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{L_s + Z / LZ_s}{CLRZs^2 + L(Z+R)s + RZ / LRZ_s}$$

Resultado

$$\frac{P_a(s)}{P_p(s)} = \frac{LR_s + RZ}{CLRZs^2 + L(Z+R)s + RZ}$$

Error estacionario: Este error se calcula cuando la entrada $P_a(s)$ es un escalon estacionario, es decir, $P_a(s) = 1/s$:

$$E(s) = \lim_{s \to 0} P_a(s)\left(1 - \frac{P_p(s)}{P_a(s)}\right) = \lim_{s \to 0} s\frac{1}{s}\left(1 - \frac{LR_s + RZ}{CLRZs^2 + L(Z+R)s + RZ}\right) = 0$$

$e(t) = 0v$

Estabilidad del sistema en Lazo abierto

Para analizar la estabilidad del sistema en Lazo abierto se deben calcular los polos

$$CLRZs^2 + L(Z+R)s + RZ = 0$$

$a = CLRZ$, $b = L(Z+R)$, $c = RZ$

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$$\lambda_{1,2} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4CLR^2Z^2}}{2CLRZ}$$

Hipotenso

$Z = 0.010$, $L = 0.001$, $R = 1.000$, $C = 3.500$

$$\frac{P_{a(s)}}{P_{p(s)}} = \frac{LR_s + RZ}{CLRZ_s^2 + L(Z+R)_s + RZ} = \frac{0.001_s + 0.01}{3.5\times10^{-5}{}_s^2 + 0.0010_s + 0.01}$$

$\lambda_1 = -14.429 + 8.8051i$,

$\lambda_2 = -14.429 - 8.8051i$

Por lo tanto, el sistema presenta una respuesta estable y subamortiguada

Normotenso

$Z = 0.010$ $L = 0.001$ $R = 1.000$ $C = 1.500$

$$\frac{P_{a(s)}}{P_{p(s)}} = \frac{LR_s + RZ}{CLRZ_s^2 + L(Z+R)_s + RZ} = \frac{0.001_s + 0.01}{1.5\times10^{-5}{}_s^2 + 0.0010_s + 0.01}$$

$\lambda_1 = -12.062$,

$\lambda_2 = -55.272$,

Hipertenso

$Z = 0.010$, $L = 0.001$, $R = 1.000$, $C = 1.000$

$$\frac{P_{a(s)}}{P_{p(s)}} = \frac{LR_s + RZ}{CLRZ_s^2 + L(Z+R)_s + RZ} = \frac{0.001_s + 0.01}{1\times10 + 0.0010_s + 0.01}$$

$$0.0001_s^2$$

$\lambda_1 = -10.0$,

$\lambda_2 = -10.0$,

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Modelo de ecuaciones integro-diferenciales

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t) - P_b(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_b(t)]\,dt = C\frac{dP_b(t)}{dt} + \frac{P_b(t)}{R}$$

Nota: La unica variable desconocida es la salida del sistema $P_b(t)$ NO se puede despejar de terminos en los que se este derivando o integrando

$$\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_b(t)]\,dt - C\frac{dP_b(t)}{dt} = \frac{P_b(t)}{R} + \frac{P_b(t)}{Z}$$

$$P_b(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \left[\frac{P_a(t)}{Z} + \frac{1}{Z}\int [P_a(t) - P_b(t)]\,dt - C\frac{dP_b(t)}{dt}\right]$$

$$P_b(t)\left(\frac{Z+R}{RZ}\right) = \left[\frac{P_a(t)}{Z} + \frac{1}{Z}\int [P_a(t) - P_b(t)]\,dt - C\frac{dP_b(t)}{dt}\right]$$

$$P_b(t) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_b(t)]\,dt - C\frac{dP_b(t)}{dt}\right]\frac{RZ}{Z+R}$$

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